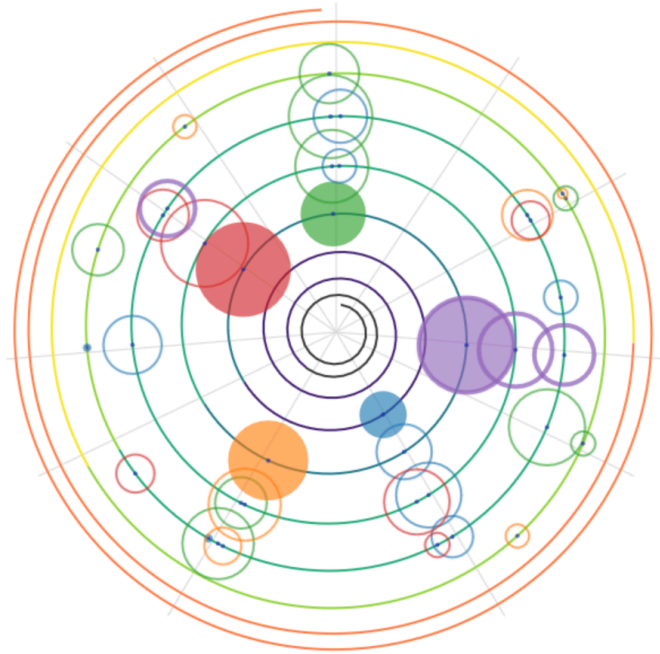




Guide to Using the Torusophone

Latest revision: May 26 2026

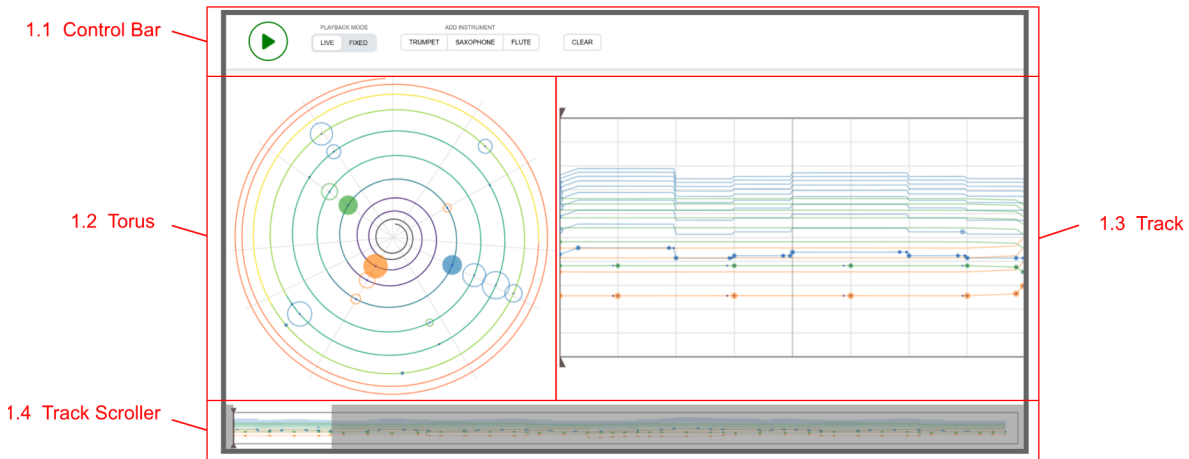
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1.0 The Torusophone Interface

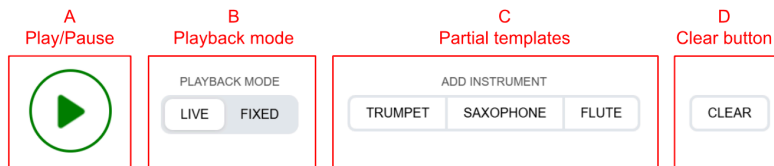
Section 1 describes the Torusophone interface. The interface has four main components.

- 1.1 Control Bar
- 1.2 Torus
- 1.3 Track
- 1.4 Track Scroller



1.1 Control Bar

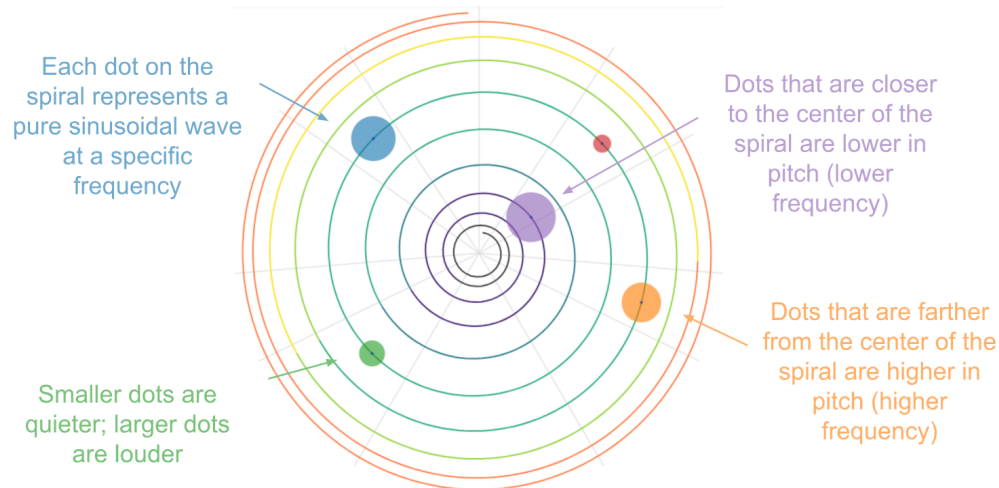
The control bar contains several controls for the user.



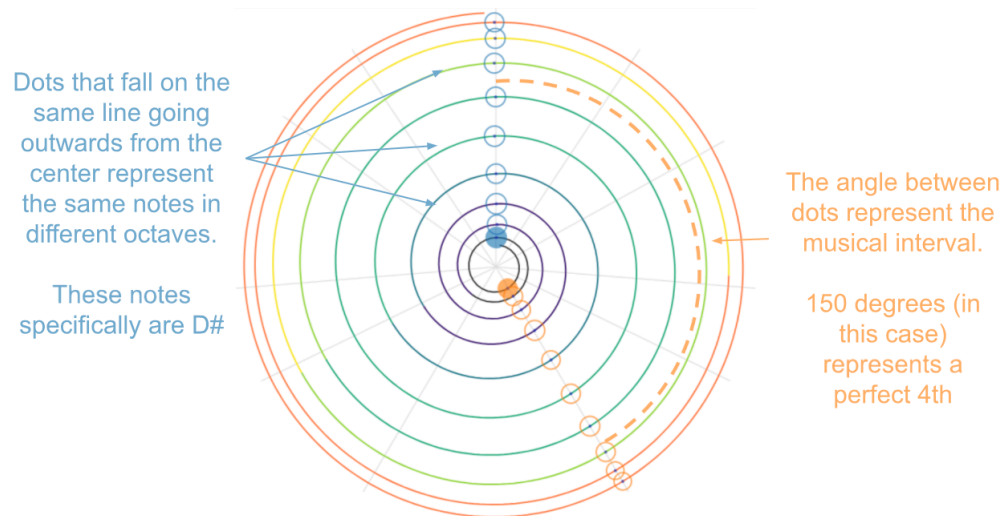
A	Play/Pause button	Allows users to listen to the track.	
B	Playback mode toggle	Toggles the playback mode. <i>LIVE</i> playback mode plays the entire track, while <i>FIXED</i> playback mode plays the notes at the current tracker time, without moving the tracker (see 1.3)	
C	Partial templates	Allows users to add preset templates of different instruments.	
D	Clear button	Removes all contents from the Torusophone interface	

1.2 Torus

The Torus provides a visualization of all partials at a particular time. The Torus is formed by a spiral with nine complete revolutions.



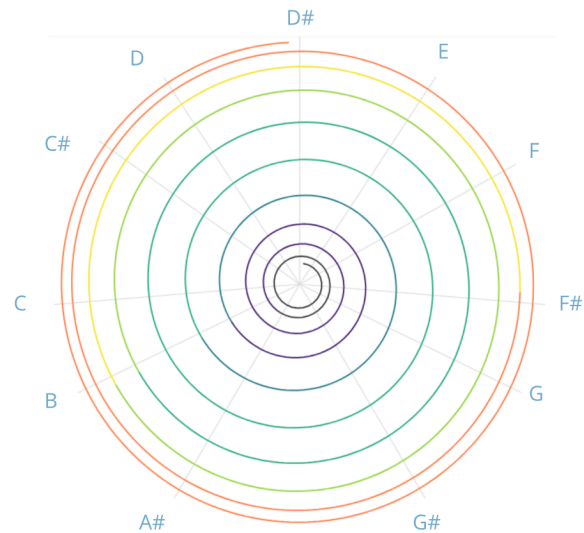
Each revolution of the spiral corresponds to a musical octave, and the music interval between two partials is defined by the angle between the two dots along the spiral.



Every note consists of a filled-in partial, representing the fundamental frequency, and zero or more unfilled partials, representing the overtones. This is just a visual distinction however, as the filled-in and unfilled partials at the same location on the spiral sound exactly the same.

NOTE: making modifications to a filled-in partial will modify the fundamental frequency as well as all associated overtones of a note, while making modifications to an unfilled partial will modify the overtone only.

On the spiral, thin gray lines originating from the center act as references for the notes found in western music, tuned using the D# just-intonation system.

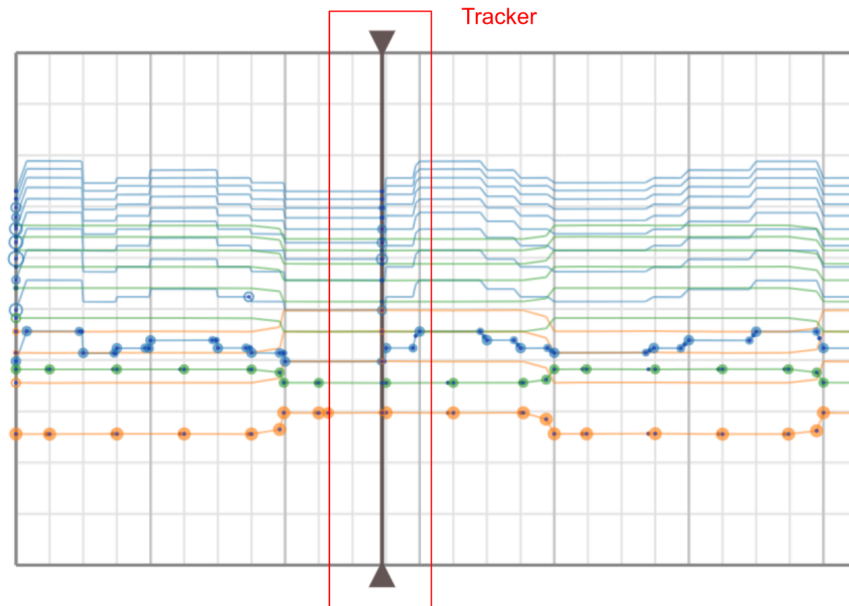


The following table shows the actions that can be taken on the Torus.

A	Selecting a partial	Click on a partial. If a filled-in partial is clicked, all associated partials (fundamental frequency and overtones) of a note will be selected.	
B	Deselecting a partial	Click anywhere not on the spiral.	
C	Adding a partial	Click anywhere on the spiral. If a partial is currently selected, an overtone will be added to the currently selected note. Otherwise, a new note will be created.	
D	Removing a partial	Hold down the <i>CTRL</i> key and click a partial. Clicking a filled-in partial will remove all associated partials of the note.	
E	Modify the frequency of a partial	Click and hold a partial while dragging it along the spiral.	
F	Modify the amplitude of a partial	Select a partial. Hold down the <i>SHIFT</i> key. Meanwhile, hold down the left mouse key anywhere on the screen, and drag the mouse toward and away from the center of the partial.	

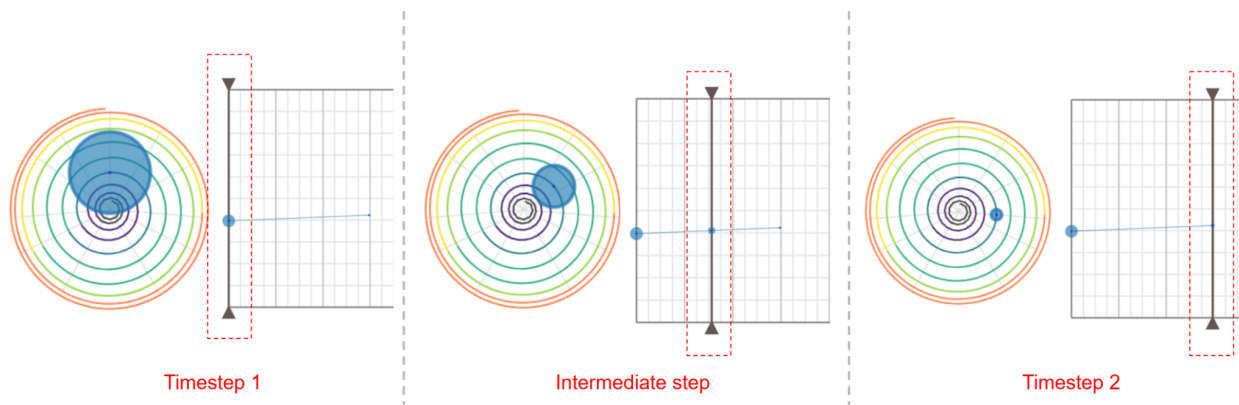
1.3 Track

The Track allows users to specify how partials change over time. Frequency is graphed on the y-axis, while time is graphed on the x-axis. A tracker specifies the current time displayed on the Torus.

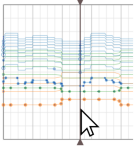
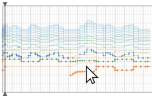
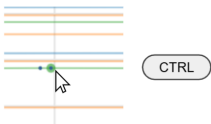
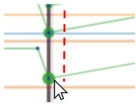
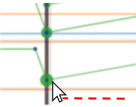


The coloured lines represent the paths that partials take across time, while coloured dots represent “timesteps” where the user has modified partials to have a specific amplitude or frequency.

The amplitudes and frequencies of partials at intermediate steps are determined through linear interpolation, like shown below.



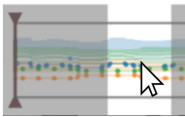
The following table shows the actions that can be taken on the Track.

A	Moving the tracker	Click anywhere on the Track. The tracker will jump to the location where the track was clicked. Alternatively, drag the mouse while holding the left mouse button to move the tracker.	
B	Zooming in/out of the time axis	Use the mouse wheel to scroll, while hovering over the track.	
C	Removing partials	Hold down the <i>CTRL</i> key and click a partial. Clicking a coloured dot will remove the note/partial timestep, while clicking a coloured line will remove the entire note/partial path up until the tracker location.	
D	Modifying the frequency of a partial	Click and hold a partial dot or line while dragging it vertically.	
E	Modifying the time of a partial	Click and hold a partial dot or line while dragging it horizontally. <i>WARNING: if a partial timestep is dragged over an existing timestep for the same partial, it will overwrite the existing timestep</i>	

1.4 Track Scroller

The track scroller shows the entire track, and allows the location of the track viewport, and tracker, in relation to the entire track.

The following table shows the actions that can be taken on the Track scroller.

A	Moving the track viewport	Click anywhere on the Track scroller. The track viewport will jump to the location where the track scroller was. Alternatively, drag the mouse while holding the left mouse button to move the track viewport.	
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2.0 Additional Controls

The following table shows additional controls on the Torusophone interface.

A	Undo	Press <i>CTRL+Z</i> to undo the previous action
B	Redo	Press <i>CTRL+Y</i> to revert the previous undo action